ASR and VITS

A practical guide to speech recognition, synthesis and evaluation

1. What does an ASR model do?

Automatic Speech Recognition (ASR) converts speech into text. Typical applications include captions, meeting transcription and voice input. VITS works in the opposite direction: it converts text into speech.

A typical ASR pipeline is **audio waveform → acoustic features / audio encoder → sequence recognition → text**. Architectures vary. Whisper, for example, uses log-Mel spectrograms and an encoder-decoder Transformer to predict text tokens [2].

For this project: use a pretrained ASR model as a frozen, external evaluator. It does not need to be trained jointly with VITS, and its predictions do not backpropagate into VITS.

2. A compact evaluation toolkit

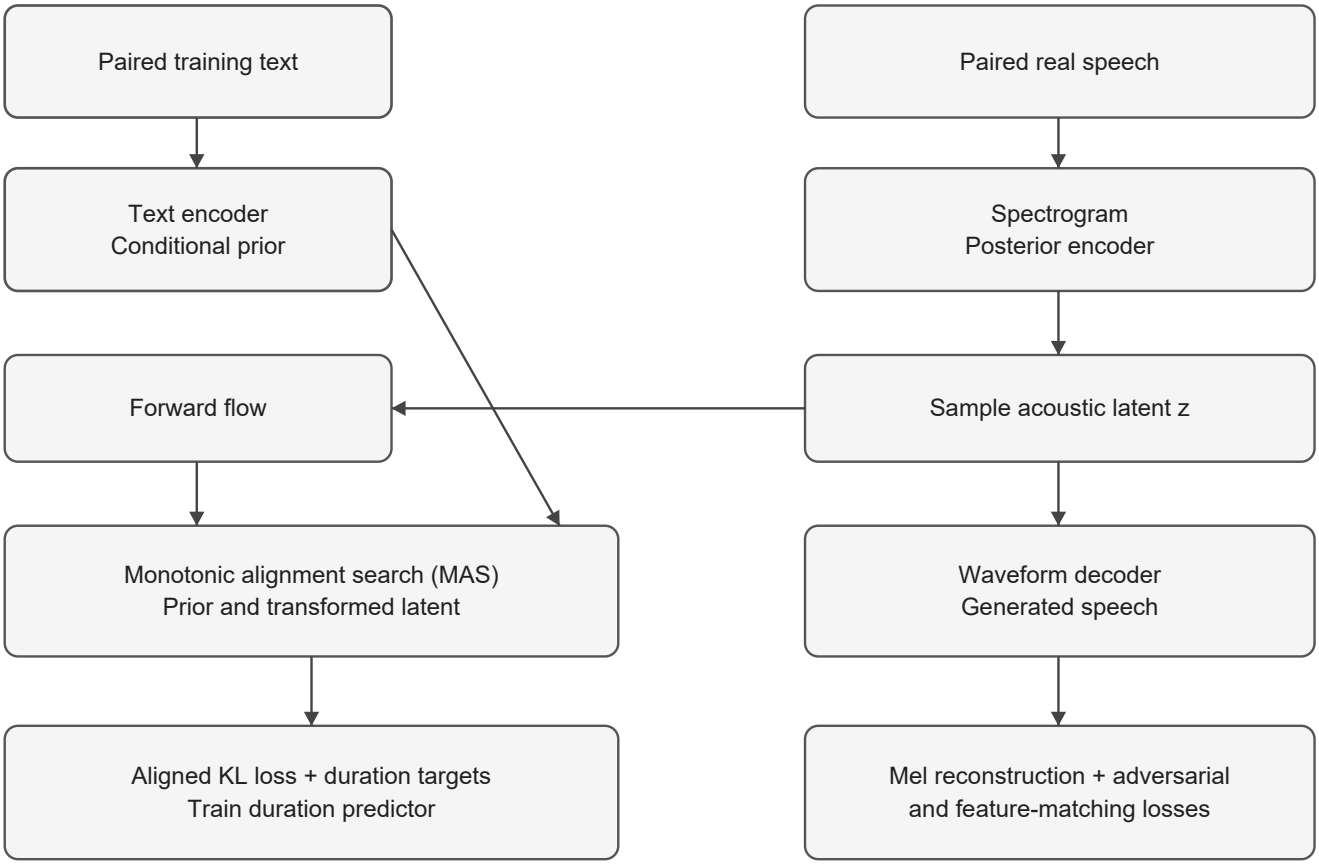
Metric	Method and interpretation
WER / CER	Transcribe generated speech and align it with the reference text. Word error rate (WER) suits word-based evaluation; character error rate (CER) is useful for Chinese. Lower is better. JiWER provides scoring and alignment tools [5].
Speaker similarity	Use an independent speaker encoder, such as ECAPA-TDNN [6], on generated and target-reference recordings. Compute embedding cosine similarity. Higher usually means more similar; it is not a probability of successful cloning.
Human MOS	Run a randomized, blinded listening test with a consistent 1-5 naturalness scale. Report the mean and uncertainty. Ask about speaker similarity separately, with a target reference.
UTMOS	A pretrained model predicts perceived speech quality [4]. Label the result as predicted MOS. Use it to supplement, not replace, human listening.
Real-time factor	RTF = synthesis time / generated audio duration. Lower is faster; RTF below 1 means generation is faster than playback. Report hardware and timing boundaries.

WER = (S + D + I) / N. S: substitutions; D: deletions; I: insertions; N: reference words. CER applies the same idea to characters. WER can exceed 100%.

Training and application

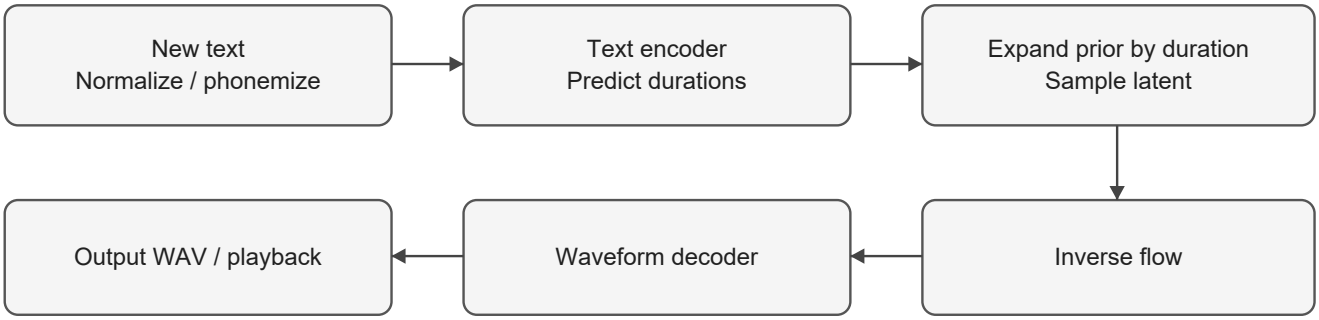
Two different paths through VITS [1] | Simplified conceptual diagrams

3. Training roadmap



The flow maps between latent spaces; it does not concatenate two latents. MAS aligns text units with audio frames. Duration targets train the duration predictor. Reconstruction compares generated and real speech; adversarial training uses both. Speaker conditioning is omitted here for clarity.

4. Application roadmap: trained TTS inference

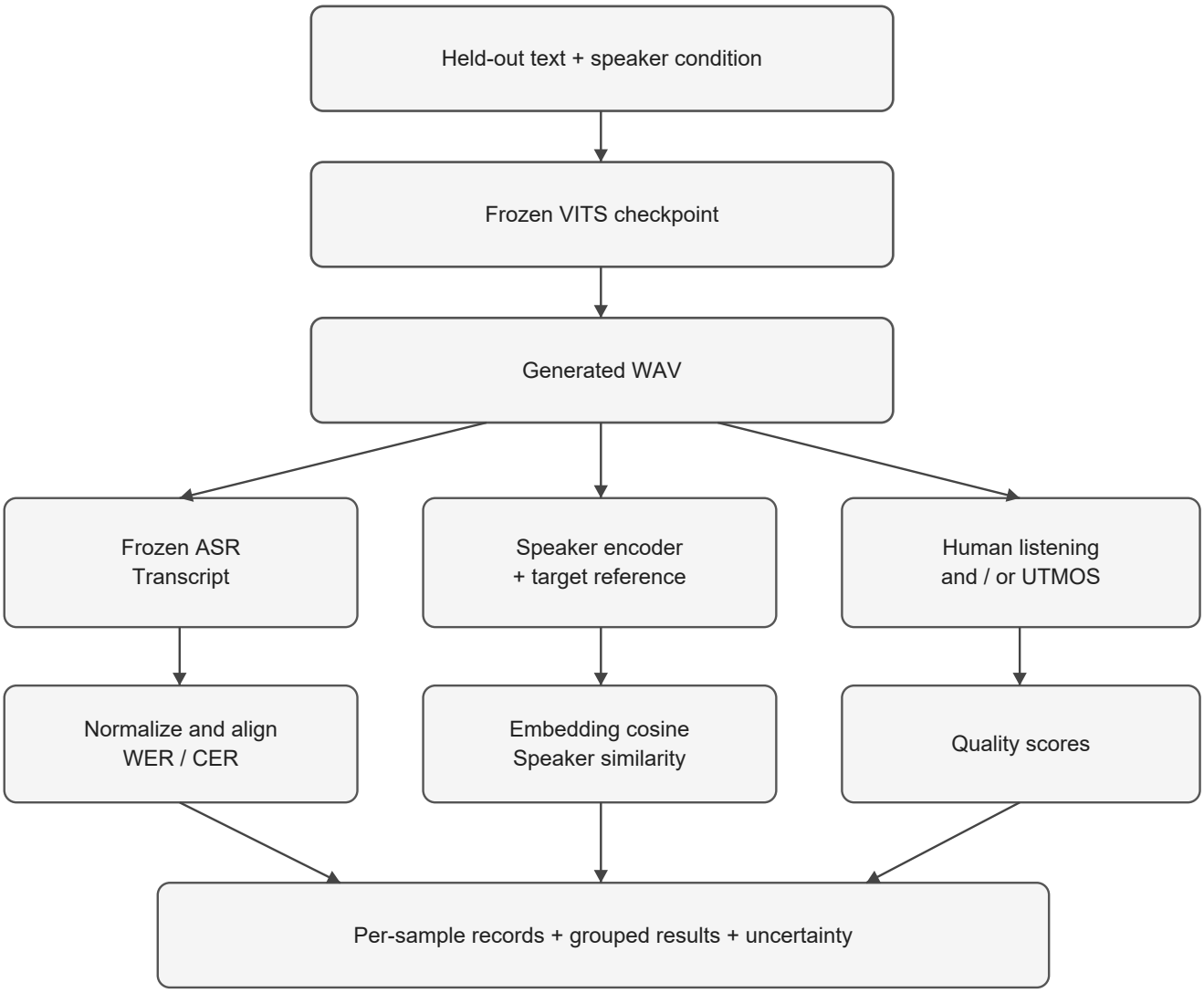


For multi-speaker VITS, a learned speaker ID conditions the relevant modules. No paired recording or discriminator is required at inference. Arbitrary unseen-speaker cloning from a reference clip requires additional support, such as a YourTTS-style extension [3].

Evaluation in practice

Keep the generator and evaluators fixed while comparing experimental conditions

5. Evaluation roadmap



The original text supplies the scoring reference. Separately, record synthesis time and WAV duration for RTF; exclude ASR and evaluation time from synthesis timing.

6. Implementation checklist

1. Generate audio from a fixed test manifest. Save the checkpoint, speaker ID, text, seed and inference settings.
2. Resample an evaluation copy to each evaluator's required sample rate and channel format. Keep original WAV files.
3. Run ASR in transcription mode, using fixed language and decoding settings. Never provide the correct transcript as an ASR prompt.
4. Apply the same predefined text normalization to reference and hypothesis. Save alignments and substitutions, deletions and insertions.
5. Score identity and quality, measure RTF, then export per-item results and aggregate statistics.

Experiment design and reading

A small, reproducible evaluation study is a useful project contribution

7. Make the comparison trustworthy

Check the evaluator. Pass real recordings through the same ASR to establish a reference level. ASR can make mistakes or infer unclear words from context. A low WER does not prove naturalness or target-speaker identity; subtracting real-audio WER does not isolate pure VITS error.

Use controlled comparisons. Include ordinary sentences, long sentences, numbers and abbreviations. Compare baseline and improved preprocessing on the same items and paired seeds. Establish the intended spoken reference independently of each system's preprocessing.

Report uncertainty. Select settings on validation data and reserve test data for final reporting. Report paired changes and bootstrap confidence intervals; account for speaker clustering in multi-speaker data. Record failure cases as well as averages.

Suggested question: Does targeted text normalization reduce content errors without reducing speaker similarity or perceived naturalness? First check what the existing frontend already handles. This is an inference-pipeline improvement, not a new model architecture.

8. References and reusable tools

[1] Kim, Kong & Son (2021). Conditional Variational Autoencoder with Adversarial Learning for End-to-End Text-to-Speech. ICML. Architecture, training and inference.
https://proceedings.mlr.press/v139/kim21f.html

[2] Radford et al. (2023). Robust Speech Recognition via Large-Scale Weak Supervision. ICML; preprint 2022. Whisper ASR foundations.
https://arxiv.org/abs/2212.04356

[3] Casanova et al. (2022). YourTTS: Towards Zero-Shot Multi-Speaker TTS and Zero-Shot Voice Conversion for Everyone. ICML. Voice similarity and listening-test design.
https://proceedings.mlr.press/v162/casanova22a.html

[4] Saeki et al. (2022). UTMOS: UTokyo-SaruLab System for VoiceMOS Challenge 2022. Automatic MOS prediction.
https://arxiv.org/abs/2204.02152

[5] JiWER. Reusable WER / CER scoring and text-alignment tooling.
https://jitsi.github.io/jiwer/

[6] SpeechBrain ECAPA-TDNN pretrained speaker encoder.
https://huggingface.co/speechbrain/spkrec-ecapa-voxceleb

9. Converting speaker A into speaker B

Cascade: A's speech → ASR → text → B-conditioned TTS. Easy to connect, but timing, emotion and pauses may change, and ASR errors propagate.
Direct voice conversion: A's speech → content representation + B's speaker condition → waveform. The model must preserve content while replacing speaker identity. Multi-speaker VITS and YourTTS offer relevant starting points [1, 3].